

Constrained Optimization

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In the previous unit , we discussed the maximum margin classifier. i.e.,

$$\min_{\mathbf{w}, b} \quad \frac{1}{2} ||\mathbf{w}'||^2 \quad (1)$$

subject to:

$$y_i(\mathbf{w}' \cdot x_i + b) \geq 1, \quad i = 1, \dots, n$$

This is known as optimization problem with the convex quadratic objective function and linear constraints. Also known as quadratic programming problem. If you are unfamiliar with these term then dont worry we will discuss one by one.

Quadratic programming problem is the type of non linear programming problem. It is the process of solving mathematical optimization problem by maximizing or minimizing quadratic objective function with linear constraints.

Alright, Now we are introducing lagrange multiplier to optimal margin classifier and trying to minimize the $\mathbf{w}, b, \alpha$.

Lagrange multiplier finds the local maxima or minima of a function subject to both equality and inequality constraints. It combines the two equations in one. That means, it is the weighted sum of objective and constraint function. For a few time, let's forgot about optimal margin classifier's equations and see some of the examples for solving constrained optimization problems of differentiable functions.

Example 1: Lagrange Multiplier- Equality Constraints

$$\min_{x,y} \quad f(x, y) = 2 - x^2 - 2y^2$$

subject to:

$$h(x, y) = x + y - 1 = 0$$

Here, we have the objective function $f(x, y)$ and the equality constraint as $h(x, y)$. The lagrangian expression can be defined as

$$\begin{aligned} L(x, y, \beta) &= f(x, y) + \beta h(x, y) \\ &= 2 - x^2 - 2y^2 + \beta(x + y - 1) \end{aligned}$$

Example 2: Lagrange Multipliers- Multiple Constraints

$$\min_{x,y} \quad f(x, y) = x^2 + y^2$$

subject to:

$$\begin{aligned} h_1(x, y) &= x + y = 0 \\ h_2(x, y) &= y + 1 = 0 \end{aligned}$$

Here, we have the objective function $f(x, y)$ and two equality constraint as $h_1(x, y)$ and $h_2(x, y)$. Then, the lagrangian expression can be defined by

$$L(x, y, \beta_1, \beta_2) = f(x, y) - \sum_i \beta_i h_i(x, y)$$

$$L(x, y, \beta_1, \beta_2) = (x^2 + y^2) - \beta_1(x + y) - \beta_2(y + 1)$$

Example 3: Primal Problem- Equality and Inequality Constraints

$$\min_{x,y} \quad f(x, y) = x^3 + y^2$$

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subject to:

$$g(x, y) = x^2 - 1 \leq 0$$

$$h(x, y) = x + y = 0$$

Here, we have the objective function $f(x, y)$ and both the equality and inequality constraint as $h(x, y)$ and $g(x, y)$. Then, the Lagrangian expression can be defined by

$$L(x, y, \alpha, \beta) = f(x, y) - \sum_i \alpha_i g_i(x, y) - \sum_i \beta_i h_i(x, y)$$

Where α is the Lagrange multiplier for inequality constraint and β for equality constraint. This is also known as generalized Lagrangian. Also, we have certain conditions as KKT-Kuhn-Tucker (KKT) to find the optimal solution in non-linear programming. It also helps to find the maximum or minimum of a function subject to some constraints.

KKT Conditions:

1. Lagrangian Stationarity

$$\nabla L(x^*, y^*, \alpha^*, \beta^*) = 0$$

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2. Primal Feasibility

$$g(x^*, y^*) \leq 0$$

$$h(x^*, y^*) = 0$$

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3. Dual Feasibility

$$\alpha \geq 0$$

4. Complementary Slackness

$$\alpha \cdot g(x^*, y^*) = 0$$

Note: * denotes the optimal value of each parameter.

Alright! let's back to our equation (1) of optimal margin classifier.

$$\min_{\mathbf{w}, b} \quad \frac{1}{2} \|\mathbf{w}\|^2$$

subject to:

$$y_i(\mathbf{w}^T \cdot \mathbf{x}_i + b) \geq 1, \quad i = 1, \dots, n$$

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We will multiply by -1 to change the $\geq$ inequality sign to $\leq$

$$-y_i(\mathbf{w}^T \cdot \mathbf{x}_i + b) \leq -1, \quad i = 1, \dots, n$$

$$-y_i(\mathbf{w}^T \cdot \mathbf{x}_i + b) + 1 \leq 0, \quad i = 1, \dots, n$$

According to the KKT complementarity conditions, we will have $\alpha_i > 0$ only for those data points that lies in the marginal line.(which is equal to value 1)

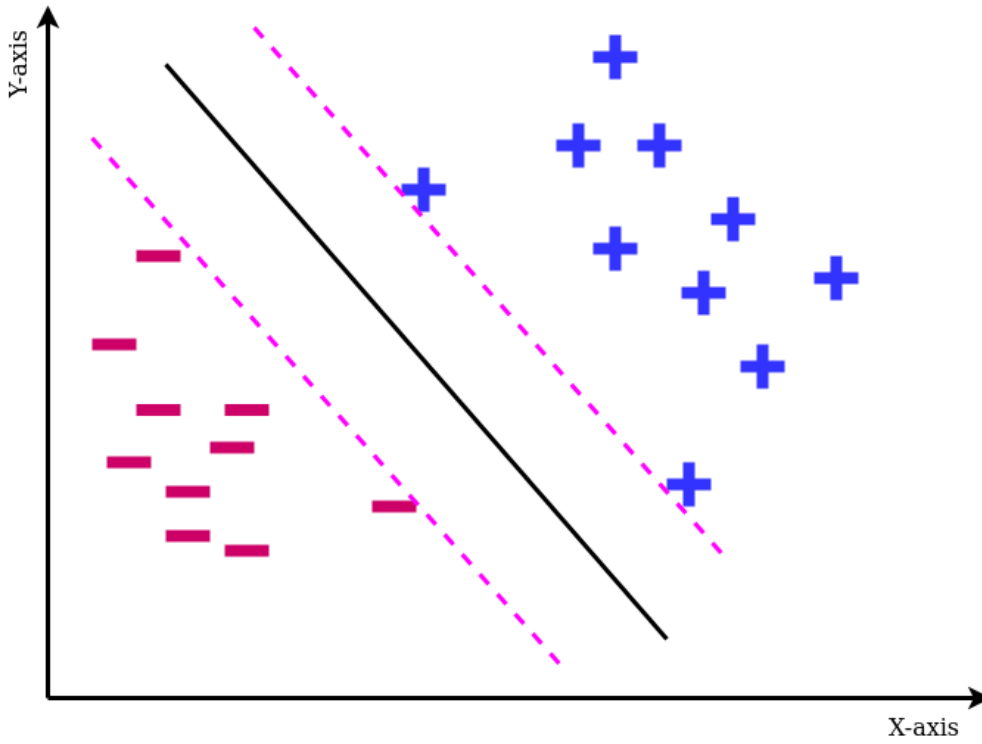


Figure 1: Representation of support vectors

In the above figure, we have three data points (one negative and two positive) that lie on the dashed line which are parallel to the decision boundary. Thus, these three α_i points will have non-zero values. These three points are called **support vectors**.

By the definition, Support vectors are the data points that are closest to the decision boundary or separating hyperplane.

Note: The support vectors can be of any number but obviously less than the training data points.

Applying Lagrangian for our optimization problem (1), we will get

$$L(\mathbf{w}, b, \alpha) = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^n \alpha_i [y_i (\mathbf{w}^T \cdot x_i + b) - 1] \quad (2)$$

subject to:

$$\alpha_i \leq 0$$

Now we will find the dual form of the problem. For that we need to minimize the $L(\mathbf{w}, b, \alpha)$ with respect to $\mathbf{w}$ and b . The partial derivative is set to 0.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \mathbf{w}} &= 0 \\ \frac{\partial \mathcal{L}}{\partial b} &= 0 \end{aligned}$$

Now solving with respect to $\mathbf{w}$,

$$\begin{aligned} \frac{\partial L}{\partial \mathbf{w}} &= 0 \\ \frac{\partial}{\partial \mathbf{w}} \left(\frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^n \alpha_i (y_i (\mathbf{w}^T \cdot x_i + b) + 1) \right) &= 0 \end{aligned}$$

$$\begin{aligned}
\frac{1}{2}2\mathbf{w} - \sum_{i=1}^n \alpha_i y_i x_i - 0 + 0 &= 0 \\
\mathbf{w} - \sum_{i=1}^n \alpha_i y_i x_i &= 0 \\
\mathbf{w} &= \sum_{i=1}^n \alpha_i y_i x_i
\end{aligned} \tag{3}$$

Similarly, solving with respect to b ,

$$\begin{aligned}
\frac{\partial L}{\partial b} &= 0 \\
\frac{\partial}{\partial b} \left(\frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^n \alpha_i (y_i (\mathbf{w}^T \cdot x_i + b) + 1) \right) &= 0 \\
0 - 0 - \sum_{i=1}^n \alpha_i y_i + 0 &= 0 \\
\sum_{i=1}^n \alpha_i y_i &= 0
\end{aligned} \tag{4}$$

Substituting the equation (3) and (4) in equation (2)

$$\begin{aligned}
L(\mathbf{w}, b, \alpha) &= \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^n \alpha_i (y_i (\mathbf{w}^T \cdot x_i + b) + 1) \\
&= \frac{1}{2} (\mathbf{w}^T \mathbf{w}) - \sum_{i=1}^n \alpha_i (y_i \mathbf{w}^T \cdot x_i) - b \sum_{i=1}^n \alpha_i y_i + \sum_{i=1}^n \alpha_i \\
&= \frac{1}{2} \sum_{i=1}^n \alpha_i y_i x_i \mathbf{w}^T - \sum_{i=1}^n \alpha_i y_i x_i \mathbf{w}^T - b \sum_{i=1}^n \alpha_i y_i + \sum_{i=1}^n \alpha_i \\
&= -\frac{1}{2} \sum_{i=1}^n \alpha_i y_i x_i \mathbf{w}^T - b \sum_{i=1}^n \alpha_i y_i + \sum_{i=1}^n \alpha_i \\
&= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \alpha_i \alpha_j y_i y_j (x_i^T x_j) - 0 \\
L(\mathbf{w}, b, \alpha) &= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \alpha_i \alpha_j y_i y_j (x_i^T x_j)
\end{aligned}$$

Now putting together with the constraints $\alpha_i \geq 0$ and the constraints $\sum_{i=1}^n \alpha_i y_i = 0$. We can define the dual problem by interchanging the min to max.

$$\max_{\alpha} \quad \mathbf{w}(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

subject to:

$$\begin{aligned}
\alpha_i &\geq 0, \quad i = 1, \dots, m \\
\sum_{i=1}^n \alpha_i y_i &= 0
\end{aligned}$$

If we can solve the dual version of SVM and get the solution α^* , we can just use $\sum_{i=1}^n \alpha_i y_i x_i$ to get the solution $\mathbf{w}^*$, and it is not hard to get the solution b^* . The optimal value for the intercept term b is computed as,

$$b^* = -\frac{\max_{i:y_i=-1} \mathbf{w}^{*T} x_i + \min_{i:y_i=1} \mathbf{w}^{*T} x_i}{2}$$

By far, we now have the solution α^* , $\mathbf{w}^*$, b^* for the dual problem, and because Lagrangian of SVM satisfies all KKT Conditions, α^* , $\mathbf{w}^*$, b^* is also the solution for our primal problem, therefore, SVM optimization is solved.

Now, Imagine we have to predict the new data point z . then,

$$\begin{aligned} \mathbf{w}^T z + b &= \left(\sum_{i=1}^n \alpha_i y_i x_i \right)^T z + b \\ &= \sum_{i=1}^n \alpha_i y_i \langle x_i, z \rangle + b \end{aligned} \quad (5)$$

Hence, Our prediction only depends on the inner product of input feature vectors. In the next chapter, we will discuss this property through Kernels in our classification problem.

Key Takeaways

- The optimal margin classifier is also known as optimization problem with the convex quadratic objective function.

$$\min_{\mathbf{w}, b} \quad \frac{1}{2} ||\mathbf{w}'||^2 \quad (1)$$

- Lagrange multiplier finds the local maxima or minima of a function subject to both equality and inequality constraints.
- we will have $\alpha_i > 0$ only for those data points that lies in the marginal line. These data points are called as **Support vectors**.
- Support Vectors are the data points that are closest to the decision boundary or separating hyperplane.
- Our prediction only depends on the inner product of input feature vectors.

$$\mathbf{w}^T z + b = \sum_{i=1}^n \alpha_i y_i \langle x_i, z \rangle + b \quad (5)$$

where, z is the new input point.

References

- Books
 - Nello Cristianini, John Shawe-Taylor(2000), An Introduction to Support Vector Machines and other Kernel-based Learning Methods, 1st edition
 - Check unit 6.1.1, page 94 to understand the Maximal Margin Classifier.
 - Stephen Marsland, Machine Learning An Algorithmic Perspective, 2nd edition
 - Check Chapter 8, page 170 to understand the optimal separation and its constrained.

